

# Computing Publications with Major Personal Contributions

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## 1. Building an AI-native Research Ecosystem for Experimental Particle Physics: A Community Vision

Aarrestad, Thea Klæboe et al.  
[arXiv:2602.17582](#) | 2026-02-19

## 2. Application of performance portability solutions for GPUs and many-core CPUs to track reconstruction kernels

Kwok, Ka Hei Martin et al.  
[EPJ Web Conf. 295 \(2024\) 11003](#) | [arXiv:2401.14221](#) | [DOI:10.1051/epjconf/202429511003](#) | 2024-01-25

## 3. The U.S. CMS HL-LHC R D Strategic Plan

Gutsche, Oliver et al.  
[EPJ Web Conf. 295 \(2024\) 04050](#) | [arXiv:2312.00772](#) | [DOI:10.1051/epjconf/202429504050](#) | 2023-12-01

## 4. A Ceph S3 Object Data Store for HEP

Smith, Nick et al.  
[EPJ Web Conf. 295 \(2024\) 01003](#) | [arXiv:2311.16321](#) | [DOI:10.1051/epjconf/202429501003](#) | 2023-11-27

## 5. Evaluating Portable Parallelization Strategies for Heterogeneous Architectures in High Energy Physics

Atif, Mohammad et al.  
[arXiv:2306.15869](#) | 2023-06-27

## 6. Detector R D needs for the next generation $e^+e^-$ collider

Apresyan, A. et al.  
[arXiv:2306.13567](#) | 2023-06-23

## 7. Automated Network Services for Exascale Data Movement

Balcas, Justas et al.  
[EPJ Web Conf. 295 \(2024\) 01009](#) | [DOI:10.1051/epjconf/202429501009](#) | 2024

## 8. IRIS-HEP Strategic Plan for the Next Phase of Software Upgrades for HL-LHC Physics

Bockelman, Brian et al.  
[arXiv:2302.01317](#) | 2023-02-02

## 9. The Future of High Energy Physics Software and Computing

Elvira, V. Daniel et al.  
[arXiv:2210.05822](#) | [DOI:10.2172/1898754](#) | 2022-10-11

## 10. Portability: A Necessary Approach for Future Scientific Software

Bhattacharya, Meghna et al.  
[arXiv:2203.09945](#) | 2022-03-15

## 11. Coffea: Columnar Object Framework For Effective Analysis

Smith, Nicholas et al.  
[EPJ Web Conf. 245 \(2020\) 06012](#) | [arXiv:2008.12712](#) | [DOI:10.1051/epjconf/202024506012](#) | 2020-08-28

## 12. A Roadmap for HEP Software and Computing R D for the 2020s

Albrecht, Johannes et al.  
[Comput.Softw.Big Sci. 3 \(2019\) 7](#) | [arXiv:1712.06982](#) | [DOI:10.1007/s41781-018-0018-8](#) | 2017-12-18